PA consistency machines

Comparison, a verified 195-state alternative, and searches below 195 states

Research prepared for Vladimir Reshetnikov

Source snapshot

`5be8b9236df6a6d308195e62a906a173dc70d00a`

Abstract

The requested construction with fewer than 195 working states is **not established by this packet**. We compare the supplied PA constructions, validate a distinct 195-state binary machine, and document the exact obstruction in a genuine 194-state local quotient. That quotient preserves the entire source control-flow graph and has eight compatible local folds, but fails a physical instruction contract because the old scanner mistakes seven address zeros for a separator. One additional scanner state repairs the full runtime, giving $135 + 68 - 8 = 195$ working states.

For this repaired machine, the packet supplies the complete 390-row table, deterministic reconstruction, 32 local simulation cases, all 4,815 symbolic primitive contracts, and an exact 59,164-step blank initialization. A common-source argument proves its halting PA-equivalent to the supplied PA195 machine. It consequently inherits the supplied ordinary mathematical PA biconditional with Con(PA); no new end-to-end Lean or Rocq theorem is claimed. The comparison covers all 31 incoming archives and 36 selected distinct rooted machine graphs. An independently reproduced finite rotation census and several explicitly bounded searches find no smaller validated machine. None is a general state lower bound.

Validated new machine: 195 working states, two symbols.

$$\mathrm{PA} \vdash \neg\,\mathrm{Halt}(M_{195\mathrm{S8}}, \mathrm{blank}) \iff \mathrm{Con}(\mathrm{PA}).$$

The strict target, fewer than 195, remains unmet.

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1 Result, conventions, and evidence

The definitive new table is `machine_195s8/pa195_scanner8.tm`. The model is the same one used in the supplied PA195 construction: one deterministic tape extending infinitely in both directions, alphabet $\{0, 1\}$, an initially all-zero tape, and a head initially at coordinate zero. Working states are $0, \ldots, 194$, the start is 0, and the separate halt label is 195. Each working-state/read-symbol pair has exactly one transition. Every move is left or right by one cell. The halt state has no outgoing transition and is excluded from the count.

A line (q, b, w, D, n) means read b, write w, move in direction D, and enter n. All 195 controls are reachable in the finite transition graph. Graph reachability is not a claim that every control occurs on the actual blank execution.

Quantity	Supplied PA195	New M_{195S8}
Working controls	195	195
Binary transition rows	390	390
Inherited controls before AX allocation	134	135
Allocated AX reader controls	68	68
Compatible pair identifications	7	8
AX macro tokens	191	191
Primitive source sites	3,700	3,700
Primitive branch contracts	4,815	4,815
Exact blank startup, literal transitions	58,154	59,164

The new literal table has SHA-256

`df6ec639713c4ae7bb3656e59c833dd55ca6336a46bea8327eca639341b34662`.

The supplied PA195 table has SHA-256

`5c6afd87f7c4f24d9ad8e9e302cc21df14b6f2e3715d03322a75463a431d8455`.

Appendix A prints the complete new table. Native labels, dense labels, the source grammar, and exact certificates are separate machine-readable files.

1.1 Three different kinds of claim

Finite exact facts include table hashes, state counts, graph reachability, the 32 bounded local traces, and the complete startup tape. They are independently executable checks on fixed data.

Uniform physical correctness includes all primitive instruction contracts for arbitrary natural register contents and all admitted gap parameters. These are symbolic identities, with unchanged register-block lemmas, rather than a collection of numerical test inputs.

Arithmetic calibration is an ordinary mathematical proof that PA proves the desired consistency biconditional. Here it is inherited from the supplied PA195 report by the unconditional halting equivalence proved in Section 5. The included checkers are not a proof assistant and do not turn this ordinary proof into a newly compiled Lean or Rocq theorem.

2 Comparison with the supplied constructions

The source snapshot is ProveIt commit `5be8b9236df6a6d308195e62a906a173dc70d00a`. There are *31 ZIP archives*, not 32: the thirty-second directory item is `README.md`. The PA195 report used an older 27-archive snapshot, `906367250a7f96c923147b10df392294bced7be5`. Every one of those 27 archived byte strings is unchanged at the new snapshot. The additions are PA195 itself and three comparison/research packages; the three latter packages still target fewer than 196 states and report no success.

2.1 The useful distinctions

The original nine U208 presentations represent two rooted transition graphs, not nine independent deletions. Eighteen later helper alternatives expand that family to twenty rooted graphs. Their final saving is the same history-sensitive identification of the native 172/211 roles; adding the stated savings across those papers would count the same deletion more than once. James Chang's historical PA209 is a different parent from the native U209s of this family.

Published PA196 replaces 75 native controls of Chang's PA209 by 62 new ones, for a net reduction of 13. Its important source changes are the transposed Cantor-pair quantifier code, omission of the old seventh equality head, guarded equality vacuity, and a reorganized axiom emitter. The U208 roles 172 and 211 are already absent. The six contextual folds in the published PA196 count are also already included in its 196 states.

The strongest source simplification relevant here is the diagonal-vacuity calculus. Its 191-token emitter has 3,700 primitive sites, compared with 201 tokens and 3,812 sites for PA196. The separate-head version has 190 tokens and 3,673 sites, but the extra stack entry changes the command compiler. PA195 keeps the diagonal emitter and changes its binary association, obtaining $134 + 68 - 7 = 195$ working controls. Consequently a further useful association can preserve the existing logical calibration while changing which physical reader controls can share a history-sensitive implementation.

Construction or family	States	Source sites	Primitive cases
Published PA196	196	3812	4959
Diagonal196 and PA195	196 / 195	3700	4815
Separate196	196	3673	4780
Combined196 (H-only Shift, MP-in-H)	196	3788	4928
Diagonal + Combined	196	3676	4784
Separate + Combined	196	3649	4749
Diagonal + MP-in-H	196	3677	4785
Separate + MP-in-H	196	3650	4750
Separate + NEG18	197	3441	4468

A smaller source is not automatically a smaller physical machine. NEG18 implements $4\,\mathrm{pair}(t, 1)$ by two increments of register 7 followed by the existing `node64` core, using $\mathrm{pair}(t, 1) = \mathrm{pair}(0, t) + 2$. It has a complete 18-instruction source proof and a scratch-zero invariant, yet the supplied complete implementation uses 197 controls. The source-count saving loses useful physical sharing.

2.2 What was independently checked in this comparison

The delivered comparison corpus contains 36 selected literal tables: 32 pairwise nonisomorphic 196-state graphs, three 197-state graphs, and PA195. The older 25 selected 196-state tables are the PA195 packet's supported set. The seven additions comprise four Combined/zerojump grafts, two MP-in-H compositions,

and a separate-calculus closing-R113 variant. A fresh checker verifies every transition domain, graph reachability, unique halt, start label, literal hash and rooted action-preserving canonical graph. None of these 36 selections is an alias of another. This is a selected physical-table corpus, not an exhaustive census of every exploratory machine in every ZIP.

Evidence statuses remain explicit in the JSON and CSV. In particular the two new MP-in-H compositions have full physical instruction contracts but their packet does not complete the new arithmetic calibration. They are not silently promoted to proved PA-consistency constructions. Two older 196-state Shift experiments halt on blank tape after 43,563 steps and are excluded from this selected positive/experimental corpus. The six older diagnostic 195-state constructions likewise remain rejected; the PA195 packet is a later successful construction, not a rehabilitation of any of those failed tables.

2.3 Proof scope and useful prior negative results

The imported Lean declaration proves the internal direction $\mathrm{PA} \vdash \mathrm{Nonhalt}(\mathrm{PA196}) \rightarrow \mathrm{Con}(\mathrm{PA})$. Its external standard-model nonhalting proof is a different theorem. The PA195 report supplies an ordinary mathematical proof of both internal directions, using uniform PA proof translations, finite bit comprehension, a proof-carrying store invariant and positive physical simulation. A new physical descendant can inherit that ordinary theorem through a proved PA-formalizable halting equivalence. This requires no assertion that an end-to-end Lean or Rocq biconditional has already been exported.

Prior failures cover substantial but restricted families: direct one-row contractions, finite-window helper repairs, specified source rotations, parameter/head permutations, short scanner templates, and complete-instruction endpoint filters. For example, the cross-archive packet rejects 59,928,960 one-keeper-row assignments on finite complete-instruction fixtures. The later seven-head search records 43,033,542 scoring evaluations, still with minimum 196 in its searched family. Such results motivate changing association and simultaneous helper allocation; none is a lower bound on all smaller machines.

3 Why the 194-state local quotient is rejected

The strongest new unsuccessful candidate keeps the exact 191-token AX emitter, all 3,700 ordered primitive instructions, every ordinal successor, and the same initial source site 25. Its unfolded machine has 202 working controls. Eight disjoint, simultaneously compatible folds produce a literal 194-state, 388-row table. An independent parser, compiler, and two-cell executor confirm these finite facts.

Those facts do not establish that the physical machine executes the source. Its binary association creates seven consecutive zero bits in some source addresses. The inherited scanner counts seven zeros as a separator. It can therefore return to the source root at the wrong position. The four offending decrement sites are:

Source ordinal	Operation	Exact address
1003	decrement logical register 1	10101001101010010111000000
2310	decrement logical register 1	1010100110101100101000000
2552	decrement logical register 1	101010011010110100000
3390	decrement logical register 1	10101001101111000000

The first failed raw symbolic contract is source ordinal 1003 on the zero branch of logical register 1, which is physical register 2. The checker completes 1,309 earlier branch contracts before encountering this failure. The maximum full source address length is only 37; overall address length is not the failing condition.

The rejected table and its successful local quotient certificate are retained in `evidence/rejected_candidate_194/`. The negative replay checks the failure explicitly. They are useful together: they demonstrate why a source CFG identity and a correct physical quotient are insufficient without a valid source-to-parent runtime.

3.1 Exact scanner repair

The native left-scanning counters include $2, 3, 5, 9, 15, 23, 34$. On a one the scan resets; the old transition

$$(34, 0) \mapsto (0, L, 75)$$

recognizes the seventh zero. The repaired table inserts native control 1999:

$$\begin{aligned} (34, 0) &\mapsto (0, L, 1999), \\ (1999, 0) &\mapsto (0, L, 75), \\ (1999, 1) &\mapsto (1, L, 2). \end{aligned}$$

Every other raw row is unchanged. The identifier 1999 is fresh and is not one of the AX readers, whose labels start at 2000. The longer scan costs exactly one state.

The repaired unfolded parent has 203 controls. Applying the same eight folds gives the accepted 195-state alternative. Its maximum zero run is seven, and its entire symbolic runtime passes. Merely relaxing the zero-run filter would not have sufficed: another seven-zero grammar failed a separate contract even with this scanner. Full instruction verification remains the acceptance criterion.

4 Construction and complete physical verification

4.1 The eight simultaneous folds

All pairs below use native labels. The first state is deleted and mapped to the second:

$$\begin{aligned} &2018 \to 2024, \quad 2023 \to 2004, \quad 2030 \to 2051, \quad 2033 \to 2019, \\ &2042 \to 2028, \quad 2054 \to 2043, \quad 2056 \to 2035, \quad 2062 \to 2037. \end{aligned}$$

The final replacement transitions, including their helpers, appear in `machine_195s8/folds.json`. They are checked in the final, jointly folded table. Selecting several individually valid folds without this joint check would be unsound because a fold may change a helper used by another fold.

Lemma 4.1 (Context-sensitive quotient)**.** *Let R be the 203-state repaired parent and $Q = M_{195S8}$ its folded table. For every finitely supported initial tape, with common native start control 74, R halts if and only if Q halts. PA proves the equivalence uniformly in the finite initial-tape code.*

Proof. Let σ identify each deleted state with its keeper and fix all other states. Start 74 and halt 254 are fixed. The 25 raw transitions entering a folded pair satisfy the following history invariant: every entry to its first state writes zero and moves right; every entry to its second writes one and moves right. Thus the cell immediately left of the head records the relevant old state role. Induction on the raw run proves this invariant independently of the initial tape.

For each pair, each old role, and each current bit, the delivered local certificate runs Q from the corresponding two-cell tape and mapped control. Its endpoint has exactly the raw transition's final tape, head displacement, and mapped successor control. All intermediate reads and writes stay in these two cells, so arbitrary tape extensions are immaterial. There are 32 cases: 12 have length one, 18 length

three, and two length five. No case passes through a halt. Every other raw transition commutes with σ in a single step.

Concatenating these positive blocks simulates every raw step, with between one and five folded transitions per step. A finite raw halt therefore gives a finite folded halt. Conversely, if the folded run halted in T steps but the raw run did not halt in its first $T+1$ steps, those raw steps would yield a folded run longer than T. Determinism and the absence of halt transitions contradict that conclusion. No inverse map on arbitrary folded configurations is required.

Finite configurations and traces have primitive-recursive codes. PA proves the incoming-history invariant by induction, checks the fixed finite local identities, concatenates the bounded positive blocks, and proves the converse by the same $T+1$ argument. □

4.2 Source identity and arbitrary-register runtime

The macro emitter is unchanged as a word in the free algebra of formula constructors, reads, pushes, and guard operations. The check includes the complete resulting stack and top formula; it is not an equality test on a few numerical parameters. Two independent successor resolvers confirm the exact source graph, including all break and continue targets.

For every primitive site, the symbolic checker begins with the source path, an affine zero gap, and a complete symbolic register file. It checks both outcomes of each decrement and the exact output frame, including all registers and the updated gap. The underlying register permutation is

$$(0, 1, \ldots, 8) \longmapsto (6, 2, 0, 4, 3, 7, 8, 5, 1).$$

All 46 transition rows protecting the inherited opaque register-block lemmas are unchanged. The supplied PA195 packet contains those reference sources and their arithmetic induction arguments. Explicit sweep and register-file trace constructors, together with those induction proofs, establish the quantified premises. The finite Python execution checks the corresponding symbolic templates; it is not by itself a proof-assistant formalization of the universal contracts.

Table	Successful cases	Largest symbolic count	Sum of counts
Repaired raw parent, 203 controls	4,815	544	1,585,685
Folded M_{195S8}, 195 controls	4,815	568	1,616,840

The last two columns count operations of the symbolic executor. They are *not* uniform literal running-time bounds for arbitrary natural register values: a symbolic sweep or register-block lemma may represent an arbitrarily long finite physical excursion. The underlying physical segments are always positive and finite.

4.3 Complete blank initialization

Starting with native state 74, head coordinate zero, and an entirely zero tape, both the repaired raw table and the folded table reach the same clean source frame after exactly 59,164 literal transitions. The control is native 84, the head is at -2, and the source ordinal is 25 with path `100000`. The logical register vector is $(2, 0, 0, 0, 0, 0, 0, 0, 0)$.

The complete physical file has values

$$(0, 0, 0, 0, 0, 0, 2, 0, 0, \underbrace{1, \ldots, 1}_{\text{26 spare registers}}).$$

The header gap is 48. The certificate lists all 66 tape positions containing a one and asserts that every other position is zero. An independent dense byte-array simulator reproduces the same complete tape. Matching only a nearby window or only the visible registers would not establish this startup theorem.

5 Transfer of the internal PA biconditional

The new work can be isolated from the inherited logical calibration. Let S denote the deterministic ordinal register program whose exact 3,700 instructions and successors are shared by PA195 and M_{195S8}. Its initial ordinal is 25 and its logical register vector is as above.

Theorem 5.1 (Common-source transfer)**.** *PA proves*

$$\mathrm{Halt}(M_{195S8}, \mathrm{blank}) \iff \mathrm{Halt}(M_{195}, \mathrm{blank}).$$

This transfer uses no consistency or soundness assumption.

Proof. For either physical machine N, let $E_N(i, \mathbf{r}, g)$ denote its complete encoding of source ordinal i, register file $\mathbf{r}$, and an admitted gap g. Source paths may differ between the two grammars; ordinal instructions, register meanings, and successors do not.

Its fixed blank startup reaches E_N of the common source seed. For every nonhalting source instruction, its primitive contract gives a finite positive physical segment to the encoding of the exact source successor. For a source halt, the contract gives a finite positive segment to the distinguished physical halt. Register-file and gap preconditions are preserved. These statements hold for arbitrary natural file entries, by the symbolic identities and unchanged register-block lemmas.

PA proves by induction that a finite halting source run yields a halting physical run. For the converse, suppose a physical run halts in T transitions. It cannot halt during the certified startup. If S failed to halt through its first $T + 1$ steps, primitive recursion would generate those source steps, and their positive physical segments, with startup prepended, would yield a defined run longer than T. This contradicts determinism at the alleged halt. Hence PA proves

$$\mathrm{Halt}(N, \mathrm{blank}) \iff \mathrm{Halt}(S, \mathrm{seed})$$

for both choices of N, and the displayed equivalence follows.

The argument uses finite coded configurations, induction, and primitive-recursive generation of the finite traces and registers. It does not require an internal truth predicate for PA. □

Corollary 5.2 (Consistency calibration, ordinary mathematical level)**.** *Using the supplied PA195 calibration,*

$$\mathrm{PA} \vdash \neg\,\mathrm{Halt}(M_{195S8}, \mathrm{blank}) \iff \mathrm{Con}(\mathrm{PA}).$$

Proof. The supplied PA195 report proves the corresponding PA biconditional for M_{195} by syntactic proof translations, finite bit comprehension, a proof-carrying store invariant, and positive simulation. Combine that theorem with Theorem 5.1. □

The original report and relevant reference sources are included under `reference_pa195_packet/`. The ordinary arithmetic argument was read during this audit; this is not a claim that its entirety has been reproved by a new proof-assistant kernel run. The imported PA196 Lean endpoint proves the internal nonhalting-to- consistency direction. It must not be mistaken for a previously compiled internal biconditional for this new literal table.

6 What the searches establish

6.1 An exhaustive, finite rotation neighborhood

A move either rotates one occurrence of a binary association or rotates all occurrences of one distinct ordered subtree. Both directions $((a, b), c) \leftrightarrow (a, (b, c))$ are allowed. Macro leaves remain fixed. Every first move and every second move is enumerated; invalid intermediate encodings are retained so that a second move may repair them.

There are 218 distinct one-move results, 24,154 results obtainable in exactly two moves, and 24,319 distinct grammars in the union including the seed. These sets overlap. Of the union, 12,277 pass the stated encoding gates. Every admitted grammar is scored by exact simultaneous compatible-fold search, with finite-state cycle detection instead of a local fuel cutoff. The minimum allocator count is 195, attained by 14 union grammars.

A separate Python implementation reconstructs every canonical grammar key, the one/two-move reachability flags, the encoding properties, and every admitted fold score. It agrees with the C++ census. The complete key file, per-grammar census, sources, and receipts are included in `search/two_rotation/`. This is an exclusion for that neighborhood and allocator; it is not a claim that PA195 is minimal among all grammars or all Turing machines.

6.2 Local deletion screens and heuristic searches

On the verified 195S8 table, the two-cell screen tried 32,010 asymmetric aliases and 2,784 opposite-history pairs in both predecessor directions. Allowing arbitrary finite nonrecursive helper detours gave no deletion. A stronger two-cell screen allowed either replacement row to be revisited recursively: 5,451,830 row assignments and 29,074,603 simulated local steps also yielded no deletion. Exact cycle detection closes these finite local searches. A positive control rediscovers an existing fold in the unfolded parent.

A separate discovery screen derived conservative tape windows from the complete symbolic contracts and startup, excluding opaque and self-sweeping controls. It considered 165 controls, 13,530 pairs, and 258,720 initial actions. Both a five-cell window and a nine-cell window gave no additional deletion. This screen is documented as a particular discovery family, not as a universal no-fold theorem.

Pure control aliases, partial-provider rewrites, source rotations, equal-word subtree crossovers, scanner shortening, and several finite stochastic association searches were also tried. The receipts distinguish proposal slots, actually scored grammars, unique retained grammars, optimistic matching bounds, and genuinely compatible folds. These quantities are not interchangeable. In particular, several apparent 194 scores from plain bipartite matching disappeared when all selected folds were checked together.

The source invariant using the finite domain $\{0, 1, 2, 3, \geq 4\}$ reaches 3,697 of 3,700 sites and removes five of the 4,815 branch contracts. It exposes no free transition row in either PA195 or 195S8. It does not prove that every abstractly reachable site occurs on the concrete blank execution.

6.3 Dedicated extra routers

A distinct allocation experiment adds a read-only helper

$$(h, 0) \mapsto (0, R, u_0), \qquad (h, 1) \mapsto (1, R, u_1).$$

The helper is absent from the raw machine and appears only in the folded implementation. A correct local certificate may execute it during a detour. Its cost is worthwhile only if it enables more state deletions than the number of added helpers.

For the fixed original PA195 grammar, one-helper screening finds 1,147 useful routers and at most eight optimistic folds; nine would be needed to reach 194. For the fixed scanner-eight grammar it finds 1,287 routers and at most nine folds; ten would be needed. Both conservative incoming histories and the broader, unreachable-helper interpretation were tried. Two-router screening also checks cases where the routers complete different missing branches of the same pair. The fixed-grammar tests still give no optimistic score below 195.

Across 172 retained admissible original-PA195 grammar ties, the matching upper bound on fold savings yields an optimistic 194-state count for 24 grammars with one helper, but all fail the simultaneous compatible-fold check. The completed all-control two-helper screen also finds optimistic 194 scores on 24 grammars, and rejects all 11,655 concrete improving router assignments by exact history and cycle checks. Related AX-only and one-helper modes overlap with this search; their trial counts must not be summed as distinct machines. Across 141 distinct score-60 scanner-eight ties from two final search pools, neither one nor two added routers gives even an optimistic 194 score. Complete sources and receipts are included in `search/added_routers/`.

Finally, the start control was checked separately rather than discarded by a generic rule against merging the start. On blank tape, native 74 has a known zero on its left, consistent with its incoming history. General recursive two-cell start-fold searches over 52 eligible partners in PA195 and 51 in 195S8 both fail. This closes that specific exception to the earlier local screens, not arbitrary startup changes.

6.4 Completed association runs

Seven separately seeded root searches use 23,000,000 proposal slots: three 3,000,000-slot ordinary association runs, two 3,000,000-slot encoding-repair runs, a 3,000,000-slot scanner-eight run from the original seed, and a 5,000,000-slot scanner-eight run from the eight-fold seed. The repair runs preserve a 194-state local quotient only with remaining encoding violations. The relaxed scanner runs find no allocation below 195 after the extra scanner is counted. An additional 3,000,000-slot equal-word crossover run has the same negative outcome.

These figures count proposal slots, including proposals rejected before scoring. They are not counts of distinct machines. The separate source families in the final appendix have their own exact scoring-call totals. Finite seeds, headers, executable source code, final pools and logs are retained so that these empirical outcomes can be examined or repeated.

7 Ideas from ZFC and ConciseIPRA

The ZFC developments suggest optimizing physical allocation jointly with source syntax: predecessor markers can distinguish two logical reader roles, useful helpers can be shared, and a partial selector may use a transition row whose other role has a proved restriction. These are concrete compilation ideas rather than a change of target consistency statement. The new eight-fold parent and the additional helper searches pursue that approach.

The ZFC project also illustrates the need to keep source proofs, literal runtime checks, startup, external correctness, and internal arithmetic statements separate. Its documented K215 endpoint and the internal PA result for the distinct 240-state table have different formal scopes. Neither endpoint can simply be relabeled as a PA consistency construction [3].

ConciseIPRA provides a complementary idea: factor common proof operations, weaken a primitive rule when its full form is derivable, and combine rule families through a shared syntactic interface. Its clause, one-body, and evaluation-discharge presentations make these possibilities explicit [4]. The searches in the following appendix apply that style of factoring to the PA195 diagonal clause: move it to another

gate parameter, or bundle it with an existing negated equality-witness head.

The size measure in ConciseIPRA is the complete normalized proof checker, including significant tokens, AST nodes, and bytes. A smaller such checker does not imply a smaller binary blank-tape TM. Moreover, bi-interpretability with its stated iPRA target alone does not supply the internal PA calibration with Con(PA) required here. Both arithmetic calibration and a complete physical compiler would still be necessary before substituting that calculus.

8 Reproduction, limitations, and useful next targets

The top-level command

```
python3 verify_all.py
```

checks the package hashes and runs the principal exact finite, symbolic-runtime, rejected-candidate, comparison, and source-family checks. Use ordinary Python without `-O`; the physical reference checker deliberately refuses optimized execution. The source-family proof checks separately exercise both interpreter modes using explicit failure conditions. The two-rotation census has its own longer replay instructions; its exhaustive regeneration is not silently represented by a quick structural check.

The complete accepted machine is reconstructible with the Python standard library. A C++17 compiler is needed to repeat the finite searches. The report builds with `pdflatex` from its `report/` directory. Search executables are not required to verify the accepted table.

Three research targets remain concrete:

1. Find a physically valid 194-state association with the original scanner, or a 193-state reader/runtime allocation before paying for the eighth-zero scanner. The failed 194 seed and the accepted repair provide exact inputs and failure conditions.
2. Change a shared logical routine or macro body enough to reduce the total compiler cost. The new 51 equal-length source words offer proved source-calculus alternatives, but the searches here do not turn any into a smaller accepted machine.
3. Permit allocation changes involving more than the tested two-cell folds, one-control replacements, or bounded helper families, while retaining a fully checkable positive simulation and exact blank startup.

No empirical search in this packet proves that these approaches must fail. Conversely, no lower local score is accepted as a consistency machine until its whole chain of proof obligations passes. The requested strict reduction remains open in this delivery.

A Exact transition table of the accepted 195S8 machine

Each cell lists write symbol, direction, and next state. Working state 0 is the start; 195 is the separate halt.

State	Read 0	Read 1	State	Read 0	Read 1
0	1 R 1	1 R 61	98	0 R 91	1 R 107
1	1 R 82	1 R 1	99	0 L 131	0 L 85
2	0 L 2	1 L 3	100	0 R 108	1 R 94
3	0 L 4	1 L 3	101	0 L 109	0 L 2
4	0 L 6	1 L 3	102	0 R 38	1 R 68
5	1 R 7	0 L 5	103	0 R 59	1 R 110
6	0 L 9	1 L 3	104	0 R 59	1 R 111
7	0 R 7	0 R 10	105	0 R 66	0 L 16
8	0 R 14	1 R 13	106	0 R 14	1 R 110
9	0 L 15	1 L 3	107	0 R 21	1 R 112
10	0 L 16	0 R 10	108	0 R 113	1 L 20
11	0 L 26	1 R 18	109	0 R 30	1 R 65
12	0 R 19	1 R 20	110	0 R 38	1 L 108
13	0 L 79	1 L 31	111	0 R 121	0 L 16
14	0 L 11	0 R 22	112	0 R 114	0 L 16
15	0 L 23	1 L 3	113	0 R 24	0 R 47
16	1 L 2	0 L 2	114	0 R 48	0 R 32
17	0 R 24	0 R 32	115	0 R 70	1 R 86
18	0 R 84	0 L 95	116	0 R 19	1 R 118
19	0 R 28	1 R 29	117	0 R 126	1 R 28
20	0 R 30	1 R 31	118	0 R 122	1 R 107
21	0 R 24	0 R 57	119	0 R 94	1 R 86
22	0 R 33	1 R 22	120	0 R 83	1 R 98
23	0 L 34	1 L 3	121	0 R 87	1 R 87
24	0 L 35	0 R 1	122	0 R 17	1 L 109
25	0 R 36	1 R 25	123	0 R 70	1 R 101
26	0 R 122	1 R 41	124	0 R 96	1 R 108
27	0 R 38	0 L 39	125	1 L 39	1 L 99
28	0 R 40	1 R 41	126	0 R 80	1 R 78
29	0 R 42	1 R 43	127	0 R 120	1 R 133
30	0 R 44	1 R 45	128	0 R 124	1 R 123
31	0 R 21	1 R 46	129	0 R 123	1 R 128
32	0 R 47	1 R 32	130	0 R 44	1 R 90
33	0 R 32	1 R 33	131	0 L 5	0 L 131
34	0 L 134	1 L 3	132	0 R 127	1 R 59
35	1 R 35	1 L 2	133	0 R 11	1 R 37
36	0 R 49	1 R 36	134	0 L 69	1 L 3
37	0 R 28	1 R 20	135	0 R 136	1 R 140
38	0 L 11	0 R 36	136	0 R 12	1 R 137
39	0 L 5	1 L 51	137	0 R 138	1 R 58
40	0 R 24	0 R 25	138	0 R 86	1 R 139
41	0 R 13	0 L 16	139	0 R 78	1 L 164
42	0 R 53	1 R 54	140	0 R 141	1 R 194
43	0 R 55	1 R 56	141	0 R 64	1 R 142
44	0 R 24	0 R 52	142	0 R 143	1 R 186
45	0 R 58	1 R 59	143	0 R 12	1 R 144
46	0 R 60	0 L 16	144	0 R 145	1 R 176
47	0 R 25	1 R 47	145	0 R 146	1 R 175

Continued on next page

State	Read 0	Read 1	State	Read 0	Read 1
48	0 L 35	0 R 61	146	0 R 86	1 R 147
49	0 R 16	1 R 49	147	0 R 148	1 R 169
50	0 R 63	1 R 64	148	0 R 78	1 R 149
51	0 L 39	1 L 51	149	0 R 150	1 R 162
52	0 R 22	1 R 52	150	0 R 126	1 R 151
53	0 L 11	0 R 49	151	0 R 152	1 R 157
54	0 R 58	0 L 16	152	0 R 157	1 R 154
55	0 R 65	1 R 66	153	0 L 183	1 R 78
56	0 R 59	1 R 67	154	0 R 155	1 R 156
57	0 R 52	1 R 57	155	0 R 116	1 R 153
58	0 R 48	0 R 25	156	0 R 139	1 R 136
59	0 L 28	1 L 68	157	0 L 145	1 R 153
60	0 L 102	1 L 30	158	0 R 159	1 R 160
61	0 R 0	1 R 61	159	0 R 160	1 R 137
62	1 R 125	1 L 51	160	0 R 19	1 R 161
63	0 R 70	1 R 195	161	0 R 58	1 L 191
64	0 L 126	1 R 31	162	0 R 181	1 R 166
65	0 R 40	1 R 60	163	0 R 12	1 R 164
66	0 L 79	1 R 71	164	0 R 153	1 R 87
67	0 R 53	1 L 72	165	0 R 137	1 R 136
68	0 R 73	1 R 74	166	0 L 190	1 L 178
69	0 R 69	1 R 75	167	0 R 164	1 R 78
70	0 R 76	1 R 77	168	0 L 156	1 L 175
71	0 R 79	1 L 110	169	0 R 170	1 R 174
72	0 L 98	1 R 54	170	0 R 163	1 R 171
73	0 R 48	0 R 33	171	0 R 172	1 R 58
74	0 R 81	0 L 41	172	0 R 161	1 R 156
75	0 R 50	1 R 132	173	0 R 116	1 L 176
76	0 R 130	1 R 21	174	0 R 155	1 R 171
77	0 R 101	1 R 90	175	0 R 158	1 R 173
78	0 R 92	1 R 93	176	0 R 177	1 R 184
79	0 R 73	1 R 72	177	0 R 178	1 R 166
80	0 R 91	1 R 119	178	0 R 179	1 R 183
81	0 R 95	1 R 96	179	0 R 180	1 R 182
82	0 R 62	0 R 1	180	0 R 58	1 R 181
83	0 R 27	1 R 97	181	0 L 169	1 R 165
84	0 R 89	1 R 88	182	0 R 139	1 R 116
85	1 L 99	1 L 85	183	0 R 173	1 R 168
86	0 R 19	1 R 100	184	0 R 166	1 R 138
87	0 R 58	1 R 58	185	0 R 116	1 R 173
88	0 R 117	1 R 115	186	0 R 187	1 R 190
89	0 L 26	1 R 135	187	0 R 188	1 R 159
90	0 R 95	0 L 131	188	0 R 116	1 R 189
91	0 R 102	1 R 103	189	0 R 168	1 R 78
92	0 R 67	1 R 56	190	0 R 191	1 R 192
93	0 R 55	1 R 104	191	0 R 185	1 R 189
94	0 R 21	1 R 105	192	0 R 167	1 R 193
95	0 R 21	0 L 10	193	0 R 168	1 R 122
96	0 R 59	1 R 41	194	0 R 129	0 L 131
97	0 R 106	1 R 8			

B Archive-by-archive comparison

The current pinned snapshot contains 31 incoming ZIP archives, totalling 233,712,151 bytes and 22,500 outer file members. All 31 archives were opened; the 27 older archive hashes agree exactly with the census included in PA195. That supplied detailed comparison is retained and extended below with the PA195 packet and three later-arriving comparison packets. Counts of raw files, source instructions, grammar nodes, physical states and validated machines are kept separate.

Archive	Contribution and status
`Below_U208_Research_2026-10-29.zip`	Source-boundary guards, displaced-head quotient lemma, closed-loop pruning, and broader negative local searches. Pruning 3909 to 3904 source occurrences increases the shared-tree cost. No accepted 207.
`ConPA_below208_research_2026-10-29.zip`	Two-object obstruction to collapsing source/output cover roles, exhaustive restricted axiom-parameter permutation search, two-row and register-invariant searches. No accepted 207.
`PA_consistency_TM_research_2026-11-13.zip`	New source-only zero-run-five witness (3812 sites,159 internal nodes,depth37), combined-negation/MP3573-site157-node witness, and parameter-free residual MP. Fixed/protected physical allocations remain rejected; many bounded quotient searches are negative.
`U208_ConPA_2026-10-29.zip`	Complete U208, original class A: identify old native roles 172 and 211, with helper 217 (or 214 under alternate native labeling). The local simulation is a finite all-tape argument; full PA calibration is inherited and written, not newly kernel exported.
`U208_ConPA_Conditional_Context_2026-10-29.zip`	Same conditional-history reduction; primary and alternate presentations give the two original U208 graph classes. Four local blocks have lengths 7,1,3,1.
`U208_ConPA_Entry_Context_2026-10-29.zip`	Re-expresses the same reduction by entry guards, with primary and compatibility alternatives. Its saving is not additive with the previous packets.
`U208_ConPA_History_2026-10-29.zip`	Historical-label presentation of the same two classes and same one-state deletion; includes a failed further reduction.
`conpa_tm_comparison_and_reduction_search.zip`	Two experimental196 MP commands (result in H, omit zero push), with mathematical command arguments but only bounded physical frame evidence. Two rejected Shift layouts actually halt from blank after43563 steps.
`conpa_tm_comparison_and_search.zip`	Accepted AD-register-swap196. Two-rotation195 is rejected by 93 primitive-contract failures. Source geometry changes cannot reuse old contextual folds automatically.
`conpa_tm_study_2026-11-13.zip`	Combined196 implements H-only Shift and MP-in-H, with4928 uniform contracts; an additional rebracketed Combined196 has full runtime and startup confirmed in this audit; direct-negation197 has3556 sites. A157-node source receives a larger207-state realization.
`pa195_conpa_2026-11-13.zip`	195-state construction: unchanged 191-token diagonal AX word and 3700 primitive sites; 134 inherited controls + 68 readers - 7 simultaneous context folds. Complete written PA biconditional, all 4815 symbolic contracts, and exact 58154-step startup; no new Lean/Rocq endpoint.
`pa196_research_2026-11-13.zip`	Exact 157-node representation of the original3812-site source, down from159; seed stays at source path100000. Claimed prospective195 allocation loses a required decoder helper. Native HA/full-induction theory is a separate research direction.

Archive	Contribution and status
pa208_research_2026-10-29.zip	Eighteen additional pairwise nonisomorphic U208 helper variants, each with a local halting-equivalence argument. Expands accepted U208 corpus to 20 classes; no accepted 207.
pa_below196_research.zip	Eight equivalent gate spellings, orientation involutions, exact freshness obstruction R(5)=9, smaller pairing macro that increases DAG cost, H-only guard specialization, and broader finite physical searches. No new accepted smaller table.
pa_consistency_comparison_2026_11_13.zip	First quantifier head split into modal B plus explosion, with finite Hilbert certificates in both directions. Countermodels obstruct deleting either membership rewrite head or replacing the first quantifier head by reflexivity. Twelve source/table combinations retain separate command/calibration obligations.
pa_consistency_comparison_and_new196_2026-11-13.zip	Three accepted diagonal-calculus196 layouts: diagonal196 (191 AX items,3700 sites), separate196 (190 items,3673 sites), scanner_best196 (same3673 sites, improved address layout). Also a runtime/startup-checked scanner196 on original source and four auxiliary diagonal closing-CD variants with differential runtime proofs, trit invariants, and exact startup. Short-scanner195 and U208/PA196 hybrid startup attempts fail stated obligations.
pa_consistency_cross_archive.zip	Four new 196-state physical grafts: diagonal/separate AX with H-only Shift and MP-in-H, optionally zerojump. Full symbolic contracts and exact startup; adjusted nine/ten-entry command compiler supplied; ordinary calculus-to-PA proof inherited. No sub-196 construction.
pa_consistency_reduction.zip	Rebracketed196; H-only Shift196 with3811 sites and57922-step startup; integrated witness gate has3745 sites but best physical table203. Two optimistic195 allocations fail, one by an exact short cycle on a source frame.
pa_consistency_reduction_audit_2026-11-14.zip	Two 196-state diagonal/separate + MP-in-H compositions and two 197-state short-negation compositions. Full primitive contracts and startup; no new arithmetic calibration claimed. Expanded parameter, association, helper, and bounded scanner searches found no sub-196 result.
pa_consistency_research.zip	New158-node source and an accepted196 regrouping; the eighth scanner zero consumes the apparent one-state saving. Its seven-zero195 version misroots on a concrete source frame.
pa_consistency_research_2026-10-29.zip	Proves zero-guarded source specialization 3909 to 3904 instructions and preserves the ordinal control graph; no smaller physical table.
pa_consistency_research_2026-11-13(2).zip	New twelve-rotation196 and diagonal198. Also a command calculus halting at successful MP deriving code1; no new binary table for that command change.
pa_consistency_research_2026-11-13.zip	Diagonal vacuity B→U(B,B), exploiting formula-code freshness; AX201 to191, source3812 to3700. Initial complete implementation199. Gives detailed written reverse PA proof translation; no new full Lean endpoint.
pa_consistency_research_2026-11-14.zip	NEG18 = INC7; INC7; node64 implements negation with 18 source instructions under proved scratch-zero conditions. Full 197-state realization has 3441 source sites and 4468 contracts. Also a separate196 closing-R113 variant, 49M seven-head search slots, and a conditional ordinary PA bridge. No sub-196 result.
pa_tm_comparison_and_reductions.zip	Two accepted196 rotations. Equivalent gate reordering shortens AX201 to199 macro items and source3812 to3773, but best physical compilation198. Finite free-algebra/logic certificates and restricted grammar searches are retained.

Archive	Contribution and status
pa_tm_extended_research_2026-11-13.zip	Four distinct one-row closing-CD variants remain196, with constrained symbolic runtime and unchanged blank startup. Joint source invariants and a generalized register executor strengthen search.
pa_tm_research_2026-11-13(2).zip	New zerojump196 and an auxiliary CD3 parent (fully replayed in this audit): native27/1 changes 0 L40 to1 R21 under the source26 R6=0 invariant; startup58402. Deleting the resulting prospective helper gives a195 table with wrong source successor and R6 increment, witnessed also on the actual blank execution.
pa_tm_research_2026-11-13.zip	New alternative196 with head order (1,2,3,5,6,4), unchanged 3812-site CFG, maximum source depth 36 rather than 38. Physical Lean baseline rebuild is supplied. R6-pruned198 passes runtime but lacks calibrated blank startup.
pa_unary_power_cover.zip	A unary-matrix power-cover theory and written mutual consistency calibration with full PA. Supplies numerical proof-search interface and logical separations; no new literal TM.
u208_comparison_research_2026-10-29.zip	Exact comparison, source/encoding research and a fully specified rejected207 frame witness. Weak/exact schema equivalence is separate from the still-open runtime/calibration replacement.
u208_research_2026-10-29.zip	Both original U208 classes with symbolic runtime and startup evidence; no further state saving.

C Moving the diagonal clause within the guarded gate

Write $\bot$ for the distinguished formula code 1 and $\neg X = (X \to \bot)$. Use the numerical constructors, six retained heads, dominated propositional head, modus ponens, and closed generalization of the PA195 calibration. In particular, all the following are syntactic formula codes; the parameters range over natural numbers, including coincident labels. Put

$$\Gamma(B) = B \to U(B,B), \qquad R(a,b,c) = V(a,b,c) \to Q(a,b,c), \qquad G(a,b,c) = \neg R(a,b,c).$$

Here V, Q are exactly the PA195 formulas. The old gate and two alternatives are

$$\begin{aligned} F_c(a,b,c) &= \neg(\Gamma(c) \to R(a,b,c)), \\ F_a(a,b,c) &= \neg(\Gamma(a) \to R(a,b,c)), \\ F_b(a,b,c) &= \neg(\Gamma(b) \to R(a,b,c)). \end{aligned}$$

All three use precisely the guard $c \neq a \wedge c \neq b$. There is no condition $a \neq b$. A failed guard returns code zero, as before.

Fixed propositional proof operations. For arbitrary formulas X, Y, the common propositional fragment supplies

$$\begin{aligned} \pi_1(X,Y) &: \neg(X \to Y) \to X, \\ \pi_2(X,Y) &: \neg(X \to Y) \to \neg Y, \\ \iota(X,Y) &: X \to (\neg Y \to \neg(X \to Y)). \end{aligned}$$

These need no semantic completeness assumption. Here are finite natural deduction templates for them, translated by the existing syntactic deduction operation. For π_1, assume $f : \neg(X \to Y)$ and $n : \neg X$. Under a further assumption $x : X$, $n(x)$ is absurd, hence implies Y; discharge x to obtain $X \to Y$ and apply f. Discharging n yields $\neg\neg X$, and double-negation elimination yields X. For π_2, assume f and $y : Y$, form the constant implication $X \to Y$, apply f, and discharge y. For ι, assume $x : X$, $n : \neg Y$, and $u : X \to Y$; the two applications $u(x)$ and $n(u(x))$ give absurdity. Discharge u, n, x. Explosion, double-negation elimination, and deduction are the unchanged finite proof-producing operations in the pinned PA196 propositional infrastructure (`Logic/Deduction.lean` and `Logic/Connectives.lean`). Consequently these are fixed primitive-recursive proof templates, whose checker equations PA proves.

Uniform translation of axiom instances. Let $\mathcal{C}_c, \mathcal{C}_a, \mathcal{C}_b$ be the calculi with the respective gates and otherwise identical rules and heads. Projection by π_2 gives $G(a,b,c)$ from the gate of any one of them. Every diagonal clause is uniformly available in each calculus:

calculus	gate instance	consequence by π_1
$\mathcal{C}_c$	$F_c(B+1, B+1, B)$	$\Gamma(B)$
$\mathcal{C}_a$	$F_a(B, B, B+1)$	$\Gamma(B)$
$\mathcal{C}_b$	$F_b(B, B, B+1)$	$\Gamma(B)$.

Both guard inequalities follow from the PA theorem $B + 1 \neq B$. For clarity, the proof replacements for a guarded instance are as follows. To obtain $F_a(a,b,c)$ in $\mathcal{C}_c$, project $G(a,b,c)$ from $F_c(a,b,c)$, project $\Gamma(a)$ from $F_c(a+1, a+1, a)$, and combine by $\iota(\Gamma(a), R(a,b,c))$. Replace a by b for F_b. Conversely, in either relocated calculus, project $G(a,b,c)$ from its current gate and obtain $\Gamma(c)$ from its auxiliary gate at $(c, c, c+1)$. Apply $\iota(\Gamma(c), R(a,b,c))$ to recover $F_c(a,b,c)$.

Translate a finite derivation recursively: preserve every common head, propositional axiom, modus-ponens node, and generalization node, and replace each gate node by the displayed finite proof. The translation uses only formula-code constructors, addition of one, and finite proof-tree composition. It is primitive recursive. PA checks its output by induction on the input proof tree, using the same propositional-template correctness lemmas and the successor inequality. If T_{ij} denotes the resulting translation, then, for $i, j \in \{a, b, c\}$,

$$\mathrm{PA} \vdash \forall p \, \forall A \, \left(\mathrm{Prf}_{\mathcal{C}_i}(p, A) \longrightarrow \mathrm{Prf}_{\mathcal{C}_j}(T_{ij}(p), A)\right).$$

In particular the calculi have PA-provably equivalent refutation predicates. Combining this with the supplied PA195 calibration transfers the same Con(PA) equivalence to either relocated source calculus. No assertion of PA's own consistency is used.

The machine-command parameter range. The actual arithmetic command is not an unconstrained oracle for a triple. On a store (t, w, d, h), its parameters are

$$\mathrm{AX}(c, x, y): \quad a = h + y, \quad b = d + x,$$

and its result resets $d = h = 0$ while pushing the same nine entries. These are the equations of `PA196/Commands/Store.lean`; they remain unchanged by moving the diagonal clause. An initial $\mathrm{AX}(0, 0, 0)$ always returns a store with $d = h = 0$; AX itself has no halting branch. The original universal refutation compiler already uses this reset before its compiled proof (`PA196/Commands/Compiler.lean`, theorem `haltWord_universal`). From the reset state every triple is reachable by $\mathrm{AX}(c, b, a)$. Thus the auxiliary relocated instance $(B, B, B + 1)$ is produced with $c = B + 1, x = y = B$, and the auxiliary old instance $(B + 1, B + 1, B)$ with $c = B, x = y = B + 1$. No negative increment or assumption about an arbitrary live value of d, h is hidden in either witness. The same selector positions, head count, failed-guard behavior, and normal reset protocol apply.

Scope of the finite audit. The accompanying independent free-algebra check verifies the complete formula expression and pushed stack for all three 191-token words, rather than checking a numerical sample. An independent interval resolver agrees with the compiler on each 3,700-site source graph and its 4,815 primitive contracts. The guard suffix is unchanged. The 256 numerical range controls and eight Boolean controls in its receipt are diagnostic checks; the uniform range and proof-translation arguments are the ones above. These results justify a source-calculus search. They do not certify a smaller physical transition table, its scanner contracts, or a new kernel-checked endpoint. The four finite deduction trees (including the antecedent-exchange template) are supplied as `proof_templates.json`. The small explicit-rule checker `check_templates.py` checks their closed conclusions and a negative control, including when Python assertions are disabled. It uses only implication introduction/elimination, falsity elimination, and classical double-negation elimination. This is a concrete certificate for the fixed propositional part, not a replacement for the inherited numerical proof compiler or its arithmetization.

C.1 Further gate transformations with explicit scope

The broader search combines the three diagonal locations with the following finite changes. First,

$$\neg(X \to (Y \to Z)) \quad \longleftrightarrow \quad \neg(Y \to (X \to Z))$$

is proved by the explicit deduction template in `proof_templates.json`; the reverse direction is the same template with X, Y exchanged. This exchanges Γ and V in the two leading antecedent positions without introducing an extra negation. Second, exchanging the numerical parameters a, b throughout a

gate preserves the guarded axiom family, since $c \neq a, b$ is symmetric and the exchange is an involution. The gate selector exchanges its two input increments after the ordinary reset; no unbounded parameter search is needed.

Third, use the four separation matrices already considered by the earlier seven-head search: the antecedents in $M(a,b) \to (c \to M(c,b))$ may be exchanged, and its conjunction with $M(c,b) \to c$ may be exchanged. Each direction is a fixed propositional proof. To transport such an equivalence through the enclosing universal quantifier, generalize its closed proof and use the retained universal implication-distribution head. Implication and negation congruence are fixed propositional templates; repeating these operations transports the matrix equivalence through S, Q, and the whole gate. Thus every resulting gate has primitive-recursive proof translations to and from the gate before the matrix exchange. These are proof operations on closed derivations; no generalization under an unverified open assumption is used.

There are $3 \cdot 2 \cdot 2 \cdot 4 = 48$ distinct 67-token gate words. Their complete free-algebra expressions and stack effects were independently checked by `verify_source_family.py`, with the unchanged PA195 gate bound exactly as variant zero. All full AX words retain 191 tokens, nine pushes, the original parameter reads, and the same guard suffix. This is a finite source search family, not a physical-state lower bound; any proposed small transition table still requires compatible-fold and complete scanner, primitive-runtime, and startup verification.

D Moving the diagonal clause to the equality-witness head

The fifth retained head is already a negated formula:

$$H_5(d,h) = \exists x_d E(h,d) = \neg U(d, \neg E(h,d)).$$

Let a denote the other freely chosen AX parameter. For any fixed choice $B \in \{a, d, h\}$, replace this head by

$$H_5^B(a,d,h) = \neg(\Gamma(B) \to U(d, \neg E(h,d)))$$

and replace the bundled PA195 gate F_c by its earlier gate $G = \neg(V \to Q)$. All other heads and inference rules remain the same. This produces the conjunction $\Gamma(B) \wedge H_5(d,h)$ in the same economical negated-implication form used by the PA195 gate. The change adds eight tokens to head 5 and removes eight from the gate. The complete AX word still has 191 tokens, nine pushed entries, and exactly the same read and guard operations.

The finite templates π_1, π_2, ι from the Gamma-relocation proof give explicit primitive-recursive translations in both directions. In the new calculus, π_2 projects every old witness head from the corresponding bundled head, and π_1 yields every $\Gamma(C)$ by taking $a = d = h = C$. Thus ι combines $\Gamma(c)$ and any guarded $G(a,b,c)$ to recover the PA195 gate $F_c(a,b,c)$. Conversely, PA195 already derives every $\Gamma(C)$ using the guarded instance $(C+1, C+1, C)$. Combine it with the original witness head by ι to obtain H_5^B; obtain G by projecting the old gate. The remaining proof nodes translate identically. PA verifies these finite syntactic replacements by the same proof-tree induction, so the refutation predicates are PA-provably equivalent.

There is no hidden guard condition on this witness-head argument. The numerical AX command pushes its heads unconditionally; the condition $c \neq a, b$ controls only the top gate formula. Consequently the choice $a = d = h = C$ is valid for extracting the bundled head even though the gate fails its guard and returns zero. After the standard reset $d = h = 0$, the AX increments realize the required tuple with nonnegative inputs. The generic head selector at the existing fifth-head position selects H_5^B; the proof compiler uses the displayed projection when an old H_5 instance is needed. No extra stack slot or new command is introduced.

The independent expression checker in `verify_source_family.py` verifies each of the three complete AX words, including the saved incoming entries, all six pushed head values, the restored old gate, and the unchanged parameter registers. The search scores in this directory are discovery evidence; none of these variants is asserted to provide a certified sub-195 transition table.

D.1 Why a literal witness reuse does not shorten this emitter

A natural shortening attempt is to reuse an already emitted head as the required arbitrary diagonal clause, thereby removing the eight-token construction instead of moving it. There is a small syntactic obstruction for the current axiom families.

The numerical implication constructor is injective. Its output is congruent to zero modulo four, whereas a universal formula has tag two. Moreover $U(x, Y) > \max(x, Y)$ for every natural pair, including zero. Thus $\Gamma(B) = I(B, U(B, B))$ cannot coincide with heads 3, 4, or 6, or with the dominated propositional head: each has an implication, of tag zero, as its outer consequent. Head 5 and the unbundled gate have outer consequent 1, which likewise differs from $U(B, B) \geq 2$.

Only heads 1 and 2 require an additional check. If

$$I(\exists x_d U(d, h), h) = I(B, U(B, B)),$$

then $h = U(B, B) > B$, while $B = \exists x_d U(d, h) > h$, a contradiction. If

$$I(U(d, U(a, h)), U(a, U(d, h))) = I(B, U(B, B)),$$

then universal-constructor injectivity gives $a = B$ and hence $B = U(d, U(B, h)) > B$, again a contradiction. All inequalities are elementary PA arithmetic facts about the displayed Cantor codes. The zero-code alias of implication does not evade them, since the universal terms are positive and strictly increasing in the sense just used.

This excludes literal coincidence with the retained heads, not derivability from them and not a smaller machine by another construction. If the parameter B is restricted to formulas already known to be theorems, then $\Gamma(B)$ is already obtained by closed generalization followed by implication introduction. Such a reuse supplies no uniform replacement, by itself, for the arbitrary negative-formula instances used in the PA195 reverse-equality proof. A genuine shortening through this route therefore requires a new calibration argument or a different arithmetic emitter.

D.2 Finite searches of the new source families

The five completed runs used 373,968 fixed-association assignments and 12,200,000 stochastic proposal slots. Together they made 10,508,034 grammar-scoring calls. No score below 61 was found.

Family	Scored evaluations	Best score	Scores below 61
Head roles and order	1,228,070	61 / 65	0
Forced head changes	1,341,329	65	0
Gamma parameter	1,982,975	64	0
48 gate forms	3,965,397	64	0
Witness bundle	1,990,263	65	0

The first row distinguishes the unchanged PA195 seed (61) from changed words (65). It exhausts 373,248 head-role choices and 720 head orders at fixed association before its bounded walk. The next run forces changed-word restarts. The last three runs respectively use 240, 480, and 240 restarts of 10,000 proposal

slots. The inherited fixed part has 134 controls. An admissible effective reader score of 61 corresponds to the existing 195-state allocation. The scorer checks geometry first, uses matching as an optimistic bound, and performs exact simultaneous local fold-choice checking whenever that bound is at most 64. Scores above 64 need not be realizable state counts. Even a better locally compatible score would still require the complete physical scanner/runtime and startup checks. The JSON receipts preserve seeds, proposal slots, actual evaluations, acceptances, and elapsed times; standalone C++17 sources and input headers permit replay. These finite searches are not lower bounds.

References

[1] *PA195: a 195-state binary Turing machine for the consistency of Peano arithmetic*, supplied 13 November 2026 packet. Original archive SHA-256: `0001250c43a40920027403d748bcca36e4519a5a45e94dec98cafad4f3c8ff9f`. Included unchanged in substance under `reference_pa195_packet/`.

[2] Vladimir Reshetnikov, *ProveIt*, snapshot `5be8b9236df6a6d308195e62a906a173dc70d00a`. `https://github.com/VladimirReshetnikov/ProveIt/tree/5be8b9236df6a6d308195e62a906a173dc70d00a/Computability/BusyBeaver/Research/PA`. The incoming catalogue and selected literal tables in this packet are bound to this snapshot.

[3] *Smaller machines for Con(ZFC)*, ProveIt research overview, same pinned snapshot. `https://github.com/VladimirReshetnikov/ProveIt/blob/5be8b9236df6a6d308195e62a906a173dc70d00a/Computability/BusyBeaver/Research/ZFC/README.md`.

[4] *Concise presentations of intuitionistic primitive recursive arithmetic*, ProveIt project overview, same pinned snapshot. `https://github.com/VladimirReshetnikov/ProveIt/blob/5be8b9236df6a6d308195e62a906a173dc70d00a/Logic/PrimitiveRecursiveArithmetic/ConciseIPRA/README.md`.